

DIODES – PN JUNCTION DIODE (LECTURE NOTES)

1. Introduction to Diodes

- A diode is a two-terminal semiconductor device that allows current to flow in one direction only.
- It consists of a single PN junction.
- Terminals: Anode (P-side) and Cathode (N-side).

2. PN Junction Formation

- Formed by joining p-type and n-type semiconductors.
- Electrons diffuse from n-side to p-side and holes from p-side to n-side.
- This carrier movement is called diffusion.

3. Depletion Region

- Recombination of electrons and holes creates a charge-free region.
- This region is called the depletion layer.
- Charge neutrality condition: $D_p \times N_A = D_n \times N_D$.

4. Potential Barrier

- Electric field across depletion region creates barrier potential.
- Si: 0.6–0.7 V, Ge: 0.2–0.3 V.
- $V_B = (kT/q) \ln(N_A N_D / n_i^2)$, where $(kT/q) \approx 0.026$ V at 300 K.

5. Biasing of Diode

Forward Bias: Barrier reduces, depletion narrows, diode conducts after knee voltage. **Reverse Bias:** Barrier increases, depletion widens, very small current flows.

6. I–V Characteristics

- Forward region shows knee voltage and rapid rise in current.
- Reverse region shows saturation current and breakdown at high voltage.

7. Breakdown & Zener Diode

- Excess reverse voltage causes breakdown.
- Zener diodes use controlled breakdown for voltage regulation.

8. Effect of Temperature

- Increase in temperature increases reverse saturation current.
- Knee voltage decreases and forward current increases.

9. Applications

- Rectifiers, voltage regulators, clippers, clampers, switching circuits.

10. Summary

- Diode is a unidirectional device.
- Forward bias → conduction, Reverse bias → insulation.
- Temperature affects diode performance.